

GENERAL STRUCTURAL NOTES:

1. THE GOVERNING BUILDING CODE IS THE MINNESOTA STATE BUILDING CODE 2020 EDITION AS APPROVED AND AMENDED BY THE CITY OF ST.CLOUD, MN.
2. CONTRACT DOCUMENTS INCLUDE THE STRUCTURAL DRAWINGS AND SPECIFICATIONS, BUT DO NOT INCLUDE SHOP DRAWINGS, VENDOR DRAWINGS, OR OTHER SUBMITTALS BY THE CONTRACTOR.
3. CONTRACTOR SHALL CROSS VERIFY ALL CONTRACT DOCUMENTS, ELEVATIONS, DIMENSIONS, AND EXISTING CONDITIONS PRIOR TO STARTING WORK. DISCREPANCIES OR CONFLICTS SHALL BE NOTED TO THE EOR IMMEDIATELY FOR REMEDIATION. SPECIFIC NOTES AND DETAILS SHALL PRESIDE OVER GENERAL NOTES AND SPECIFICATIONS.
4. THE DIMENSIONS, LOCATIONS, AND DETAILS SHOWN ON THE DRAWINGS ARE BASED ON THE BEST AVAILABLE INFORMATION AT THE TIME OF THE DRAWINGS BEING ISSUED. DEVIATIONS WHICH ARE NECESSARY OR WHICH CONFLICT SHALL BE REPORTED TO THE EOR. CONTRACTOR SHALL HAVE FULL RESPONSIBILITY FOR DEVIATIONS NOT APPROVED BY THE EOR.
5. COSTS OF ADDITIONAL DESIGN WORK DUE TO THE SELECTION OF AN OPTION OR DUE TO ERRORS OR OMISSION IN CONSTRUCTION SHALL BE THE RESPONSIBILITY OF THE CONTRACTOR.
6. THE CONTRACTOR HAS SOLE RESPONSIBILITY FOR THE MEANS, METHODS, TECHNIQUES, SEQUENCES, AND PROCEDURES OF CONSTRUCTION. THE STRUCTURAL DRAWINGS REFLECT THE COMPLETED STRUCTURE, BRACING, SHORING, AND PROTECTION DURING CONSTRUCTION IS THE RESPONSIBILITY OF THE CONTRACTOR. THE STRUCTURE SHALL NOT BE LOADED WITH CONSTRUCTION MATERIALS AND EQUIPMENT THAT EXCEEDS THE DESIGN LOADS.
7. PENETRATIONS NOT SHOWN ON THE DRAWINGS MUST BE APPROVED BY THE EOR BEFORE PLACING THROUGH STRUCTURAL ELEMENTS. CONTRACTOR SHALL PROVIDE A CAST-IN SLEEVE FOR ALL HORIZONTAL ELEMENTS THAT EXTEND THROUGH FOOTING AND FOUNDATION WALL, SUCH AS DRAIN TILE, CONDUIT, PIPING, ETC. COORDINATE SLEEVES WITH EOR. SEE MECHANICAL, ELECTRICAL, AND ARCHITECTURAL DRAWINGS FOR ALL PENETRATIONS AND EMBEDDED ITEMS NOT SHOWN ON THE STRUCTURAL DRAWINGS.
8. CONTRACTOR TO VERIFY ALL WEIGHTS, LOCATIONS & DIMENSIONS OF MECH. EQUIPMENT SHOWN AND NOTIFY THE EOR OF ANY DISCREPANCIES. COORDINATE THIS INFORMATION WITH ALL NECESSARY INDIVIDUALS.
9. PERIODIC SITE OBSERVATION BY REPRESENTATIVES OF SANDMAN STRUCTURAL ENGINEERS IS SOLELY FOR THE PURPOSE OF DETERMINING IF THE WORK OF THE CONTRACTOR IS PROCEEDING IN GENERAL ACCORDANCE WITH THE STRUCTURAL CONTRACT DRAWINGS. A LIMITED SITE OBSERVATION SHOULD NOT BE CONSTRUED AS EXHAUSTIVE OR CONTINUOUS TO CHECK THE QUALITY OR QUANTITY OF THE WORK, BUT RATHER PERIODIC IN AN EFFORT TO GUARD THE OWNER AGAINST DEFECTS OR DEFICIENCIES IN THE WORK OF THE CONTRACTOR. ALL MATERIALS AND WORKMANSHIP ARE SUBJECT TO THE REVIEW OF THE EOR.
10. SEE THE FOLLOWING DETAILS FOR STANDARD DETAILS:
- CIP CONCRETE & FOUNDATION: 1/301
STRUCTURAL STEEL: 1/5430
CMU: 1/5322
WOOD FRAMING: 1/5410

SHOP DRAWING & DEFERRED DESIGN SUBMITTAL NOTES:

1. CONTRACTOR TO FURNISH COMPLETE AND DETAILED SHOP DRAWING & SUBMITTALS FOR REVIEW AND APPROVAL BY THE EOR. THE FOLLOWING ARE REQUIRED FOR THIS PROJECT.
- | SHOP DRAWING OR SUBMITTAL | NOTES/COMMENTS |
|--|-----------------------------------|
| CONCRETE MIX DESIGNS | BY 3RD PARTY TESTING AGENCY |
| CONCRETE REINFORCEMENT | - |
| STRUCTURAL & MISC. STEEL | - |
| CMU, MORTAR, & GROUT MATERIAL SPECS | - |
| CMU REINFORCEMENT | - |
| PRECAST/PRESTRESSED CONCRETE | DEFERRED SUBMITTAL: SEE NOTE #4 |
| LUMBER MATERIAL GRADE & DOVEL-TYPE FASTENERS | - |
| METAL PLATE CONNECTED WOOD TRUSSES | DEFERRED SUBMITTAL: SEE NOTE #4 |
| ALUMINUM FRAMING ASSEMBLIES | DEFERRED SUBMITTAL: SEE NOTE #4 |
| HANDRAILS AND GUARDRAILS | DEFERRED SUBMITTAL: SEE NOTE #4 |
| ELEVATOR | PROVIDE LOADS @ STRUCTURAL SYSTEM |
| SIMPSON STRONG-TIE ANCHOR TIEDOWN SYSTEM | - |

2. CONTRACT DRAWINGS SHALL NOT BE REPRODUCED FOR USE AS SHOP DRAWINGS. ALL SHOP DRAWINGS MUST BEAR THE REVIEW STAMP OF THE CONTRACTOR BEFORE THEY ARE REVIEWED BY THE EOR.
3. SHOP DRAWINGS SHALL SHOW ALL FIELD DETAILS AND ADDITIONAL INFORMATION NEEDED FOR THE CONTRACTOR TO CONSTRUCT THE BUILDING PER THE CONTRACT DOCUMENTS.
4. STRUCTURAL COMPONENTS/SYSTEMS DESIGNATED AS A "DEFERRED SUBMITTAL" OR AS "DELEGATED DESIGN", "DESIGNED BY OTHERS", OR "PRE-ENGINEERED" MUST INCLUDE A CALCULATION PACKAGE THAT IS STAMPED AND SIGNED BY A REGISTERED PROFESSIONAL ENGINEER, LICENSED IN THE STATE THE PROJECT WILL BE CONSTRUCTED, PRIOR TO SUBMITTAL FOR APPROVAL TO THE EOR.
5. SHOP DRAWINGS SHALL BE AVAILABLE ON THE JOB SITE DURING TIMES OF INSPECTION AND SHALL BE CLEARLY INDICATED THAT THEY HAVE BEEN REVIEWED AND APPROVED BY THE EOR.
6. REVIEW OF SUBMITTALS AND SHOP DRAWINGS BY THE EOR DOES NOT RELIEVE THE CONTRACTOR OF THE SOLE RESPONSIBILITY TO REVIEW AND CHECK ALL SUBMITTALS AND SHOP DRAWINGS BEFORE SUBMITTING TO THE STRUCTURAL ENGINEER. THE CONTRACTOR REMAINS SOLELY RESPONSIBLE FOR THE ERRORS AND OMISSIONS ASSOCIATED WITH THE PREPARATION OF SHOP DRAWINGS AS THEY PERTAIN TO MEMBER SIZES, DETAILS, AND DIMENSIONS SPECIFIED IN THE CONTRACT DOCUMENTS.
7. ALL HANDRAIL AND GUARDRAIL SYSTEMS SHALL BE DESIGNED TO RESIST ASCE-7 CHAPTER 4 DESIGN LOADING CRITERIA AND TO TRANSFER THESE LOADS THROUGH THE SUPPORTS TO THE STRUCTURE.

DESIGN CRITERIA/DESIGN LOADS:

SNOW LOAD DESIGN CRITERIA				
GROUND SNOW LOAD	Pg	50 PSF	-	
SNOW IMPORTANCE FACTOR	Is	1.0	-	
EXPOSURE FACTOR	Ce	1.0	-	
THERMAL FACTOR	Ct	1.1	TYPICAL	
THERMAL FACTOR	Ct	1.2	CANOPY	
SLOPED ROOF FACTOR	Cs	1.0	TYPICAL/CANOPY	
ROOF DESIGN LOADS				
LOAD TYPE	NOTATION	LOAD	NOTES	
SNOW LOAD	S	38.5 PSF	TYPICAL (Ps)	
SNOW LOAD	S	42 PSF	CANOPY (P)	
LIVE LOAD	RLL	20 PSF	-	
DEAD LOAD	D	20 PSF	D TC = 12 PSF D BC = 8 PSF	
FLOOR DESIGN LOADS				
FLOOR	DESCRIPTION	DEAD LOAD (D)	LIVE LOAD (L)	NOTES
1ST	TYP RESIDENTIAL	95 PSF*/135 PSF**	40 PSF**	-
1ST	STAIRS/LANDINGS/ CORRIDOR/ PUBLIC AREAS / COMMUNITY SPACE	95 PSF*/135 PSF**	100 PSF	-
1ST	ELECT/MECH/ TRASH/STORAGE	95 PSF*/135 PSF**	80 PSF**	-
2ND-3RD	TYP RESIDENTIAL/ CORRIDORS	26 PSF	40 PSF**	D TC = 18 PSF D BC = 8 PSF
2ND-3RD	STAIRS/LANDINGS/ PUBLIC AREAS / COMMUNITY SPACE	26 PSF	100 PSF	D TC = 18 PSF D BC = 8 PSF
2ND-3RD	ELECT/MECH/ TRASH/STORAGE	26 PSF	80 PSF**	D TC = 18 PSF D BC = 8 PSF
1ST-3RD	BALCONIES	SELF-WEIGHT	60 PSF	TOP LEVEL BALCONIES DESIGN FOR: S = 105 PSF WIND UPLIFT = 15 PSF (SERVICE)
*D BREAKDOWN (8" H.C.) = 60 PSF H.C. + 25 PSF CONCRETE TOPPING + 10 PSF S.I. **D BREAKDOWN (12" H.C.) = 100 PSF H.C. + 25 PSF CONCRETE TOPPING + 10 PSF S.I. ***DENOTES LIVE LOAD IS REDUCIBLE PER ASCE 7				
WIND LOAD DESIGN CRITERIA				
ULT. DESIGN WIND SPEED	V-ult	109 MPH	-	
NOMINAL DESIGN WIND SPEED	V-asd	84.4 MPH	-	
RISK CATEGORY	-	II	-	
EXPOSURE CATEGORY	-	C	-	
INTERNAL PRESSURE COEF.	GCPi	+/- 0.18	-	
C & C BASE PRESSURE	qh - ult	27.1 PSF	-	
END ZONE WIDTH	a	6.5 FT	-	
EQUIVALENT LATERAL EARTH PRESSURES				
SOIL	TYPE	PRESSURE	NOTES	
PER GEOTECH REPORT	AT REST	55 PCF	USED FOR BASEMENT WALL DESIGN	
REFER TO GEOTECHNICAL REPORT FOR BACKFILLING REQUIREMENTS. SOIL RETAINING/BASEMENT WALLS ON PROJECT HAVE NOT BEEN DESIGNED TO WITHSTAND HYDROSTATIC PRESSURE. PROVIDE DRAIN TILE SYSTEM & DRAIN PATH ALONG BACKSIDE OF WALL TO RELIEVE HYDROSTATIC PRESSURE. REFER TO GEOTECHNICAL REPORT FOR ADDITIONAL REQUIREMENTS.				
ESTIMATED WOOD BUILDING SHRINKAGE				
WOOD LEVEL	LEVEL SHRINKAGE	CUMULATIVE SHRINKAGE		
3RD	.164"	.648"		
2ND	.239"	.484"		
1ST	.245"	.245"		
ESTIMATED SHRINKAGE VALUES ASSUME ALL MEMBERS HAVE A MOISTURE CONTENT OF 19% AT THE TIME OF INSTALLATION AND 10% IN FINAL STATE.				
THESE VALUES HAVE BEEN PROVIDED FOR CONSIDERATION IN DESIGN AND DETAILING OF BUILDING SYSTEMS SENSATIVE TO SHRINKAGE.				
INSTALLATION OF MEP SYSTEMS ARE TO TAKE IN CONSIDERATION THE ESTIMATED SHRINKAGE OF THE WOOD STRUCTURE.				
G.C./OWNER HAS CHOSE TO USE CMU SHAFTS THROUGH WOOD STRUCTURE. DUE TO DIFFERENTIAL SHRINKAGE BETWEEN MATERIALS, FINISHES @ INTERFACE MAY REQUIRE REPETATIVE MAINTNANCE.				

FOUNDATION NOTES:

1. FOOTINGS ARE DESIGNED FOR A NET ALLOWABLE SOIL BEARING PRESSURE OF 3000PSF FOR STRIP FOOTINGS AND 3000PSF FOR PAD FOOTINGS. THE ASSUMED ALLOWABLE SOIL BEARING PRESSURE SHALL BE VERIFIED BY A LICENSED GEOTECHNICAL ENGINEER AT TIME OF CONSTRUCTION.
2. ALL FILL UPON WHICH FOUNDATION ELEMENTS BEAR OR RELY UPON FOR CONFINING RESISTANCE AND/OR LATERAL SUPPORT, SHALL BE DESIGNED, SPECIFIED, AND TESTED BY A QUALIFIED GEOTECHNICAL ENGINEER. THESE SITUATIONS INCLUDE, BUT ARE NOT LIMITED TO, SITE GRADING, OVER EXCAVATION FILL, UTILITY TRENCHES, ETC.
3. TEMPORARY AND PERMANENT SLOPE STABILITY OF UNDERLYING SOILS SUPPORTING THE CONSTRUCTION AND STRUCTURE IS THE RESPONSIBILITY OF A QUALIFIED GEOTECHNICAL ENGINEER. DESIGN OF THE ABOVE AND BELOW GRADE STRUCTURAL ELEMENTS AND SYSTEMS INDICATED ON THE STRUCTURAL DRAWINGS ASSUME THAT SLOPE STABILITY CONCERNS DO NOT EXIST. DESIGN AND SERVICEABILITY CONSIDERATION HAVE NOT BEEN INCORPORATED FOR DIFFERENTIAL VERTICAL AND/OR LATERAL DISPLACEMENT OF THE SOILS SUPPORTING THE STRUCTURE DUE TO SLOPE INSTABILITY.
4. UNLESS SPECIFICALLY PRESCRIBED IN A GEOTECHNICAL REPORT, BACKFILL ABOVE THE BOTTOM OF FOOTING ELEVATION AGAINST FOUNDATION WALL OR RETAINING WALL SYSTEMS SHALL BE COMPACTED TO A MINIMUM OF 95% OF THE STANDARD PROCTOR MAX DENSITY (ASTM D698-00A). MOISTURE CONTENT AT THE TIME OF COMPACTION SHOULD BE -1% /+3% OF OPTIMUM. GRADATION OF FILL MATERIAL SHALL HAVE LESS THAN 12% PASSING THE #200 SIEVE.
5. PROTECT FOOTING EXCAVATIONS FROM WATER, MOISTURE, OR FROST INFILTRATION. PRIOR TO PLACEMENT OF FOOTING CONCRETE, CLEAN FOOTING EXCAVATIONS OF SNOW, WATER, MUD, DIRT, AND DEBRIS. DO NOT PLACE FOOTINGS OR BACKFILL ON FROZEN SUB GRADE.
6. FROST COVER FOR FOOTINGS SHALL BE PER RECOMMENDATIONS OF THE GEOTECHNICAL REPORT. SEE PLAN FOR TOP OF FOOTING ELEVATIONS AND DETAILS FOR FOOTING STEP REQUIREMENTS.
7. THE CONTRACTOR SHALL BE RESPONSIBLE FOR NOTIFYING THE EOR IMMEDIATELY OF ANY SPECIAL SOIL OR WATER CONDITIONS THAT EXIST ON SITE.
8. BACKFILL & COMPACTION SHALL BE INSPECTED AND TESTED BY A LICENSED GEOTECHNICAL ENGINEER OR QUALIFIED FIELD TECH. THE SUBMITTAL OF TESTING REPORTS SHALL BE PER SPECIAL INSPECTION REQUIREMENTS.
9. BACKFILL SHALL BE COMPACTED BY MECHANICAL MEANS. WATER INFILTRATION SHALL NOT BE ALLOWED. BACKFILL SHALL BE PLACED IN ALTERNATING LIFTS ON EACH SIDE OF THE FDN WALLS FOR STABILITY.
10. WALL FOOTINGS ARE TO BE CENTERED ON WALLS UNO PAD FOOTINGS ARE TO BE CENTERED ON COLUMNS UNO.
11. CONTRACTOR TO COORDINATE INTERIOR & EXTERIOR TOP OF FOOTINGS WITH MEP CONTRACTORS, PRIOR TO START OF CONSTRUCTION. PLUMBING TO BE ROUTED ABOVE FOOTINGS UNLESS APPROVED BY EOR. FOOTINGS MAY NEED TO BE LOCALLY LOWERED TO ACCOUNT FOR ADJACENT PLUMBING LINES OR BASINS THAT COULD UNDERMINE SUPPORTING SOIL ALONGSIDE OR BELOW FOOTINGS.
12. IF SHOWN ON FOUNDATION PLAN, DRAIN TILE IS FOR GRAPHICAL REPRESENTATION ONLY. SIZE AND LAYOUT TO BE CONFIRMED WITH MEP CONTRACTOR AND CIVIL DRAWINGS.

CONCRETE AND STEEL REINFORCEMENT NOTES:

1. CONCRETE AND STEEL REINFORCEMENT SHALL CONFORM TO AMERICAN CONCRETE INSTITUTION (ACI) CODES AND SPECIFICATIONS, LATEST EDITION.
- ACI 301 "SPECIFICATIONS FOR STRUCTURAL CONCRETE"
ACI 315 "DETAILS & DETAILING OF CONCRETE REINFORCEMENT"
ACI 318 "BUILDING CODE REQUIREMENTS FOR REINFORCED CONCRETE"
ACI 308R "COLD WEATHER CONCRETING"
2. CAST-IN-PLACE CONCRETE COMPRESSIVE STRENGTHS REQUIRED (28 DAY):
- | | | | |
|--|-----------|-----------------|------------|
| FOOTINGS | 4000 PSI* | PIERS / COLUMNS | 4000 PSI |
| FOUNDATION WALLS | 4000 PSI* | INTERIOR SLABS | 4000 PSI |
| ABOVE GRADE WALLS | 4000 PSI* | EXTERIOR SLABS | 4500 PSI** |
| PRECAST TOPPING | 4000 PSI* | | |
| *MAX W/C RATIO OF 0.55 | | | |
| **MAX W/C RATIO OF 0.45. MIX DESIGN TO BE AIR ENTRAINED PER ACI TABLE 19.3.3.1 | | | |
3. SUBMIT CONCRETE MIX DESIGN & STRENGTH DATA TO EOR FOR APPROVAL. ALL ADMIXTURES ARE THE RESPONSIBILITY OF THE CONCRETE SUPPLIER'S ENGINEER.
4. CAST-IN-PLACE CONCRETE SHALL BE SUBJECT TO TESTING BY AN INDEPENDENT TESTING LABORATORY, SEE SPECS AND SPECIAL INSPECTION REQUIREMENTS.
5. ALL CONCRETE SHALL BE PLACED PER ACI & THOROUGHLY CONSOLIDATED BY MEANS OF A VIBRATOR, PARTICULARLY AROUND REINFORCEMENT STEEL AND CORNERS OF FORM WORK.
6. REINFORCING STEEL SHALL BE GRADE 60 DEFORMED, BILLET-STEEL, ASTM A615, UNO.
7. WELDED REINFORCING STEEL SHALL BE GRADE 60, LOW CARBON, ASTM A706, WHICH IS SPECIALLY MANUFACTURED TO BE WELDABLE.
8. WELDED WIRE FABRIC SHALL COMPLY WITH ASTM A62 AND A185 STANDARDS AND SHALL BE PLACED IN THE CENTER OF THE SLAB, UNO. LAP JOINTS A MINIMUM OF 6". EXTEND FABRIC TO BE WITHIN 1" OF SLAB EDGES.
9. PROVIDE ADEQUATE BOLSTERS, HIGH CHAIRS, SUPPORT BARS, ETC TO MAINTAIN THE SPECIFIED CLEARANCES FOR THE ENTIRE LENGTH OF ALL REINFORCING STEEL AND WELDED WIRE FABRIC.
10. PROVIDE EXTRA REINFORCEMENT AROUND ALL OPENINGS GREATER THAN 8" SQUARE OR ROUND. PROVIDE (2) #5 BARS @ 3" OC FOR EACH MAT OF BARS, AT EACH SIDE AND CORNER OF OPENING EXTENDING MINIMUM 18" PAST CORNER OF THE OPENING. PLACE 2" CLEAR FROM OPENING.
11. SEE DETAILS FOR REINFORCING LAP SPLICE SCHEDULE, UNO ON PLAN OR DETAILS.
12. CAST DOWELS, WITH STD 90 DEGREE HOOK, IN FOOTINGS FOR CONCRETE PIERS AND WALLS ABOVE. DOWELS SHALL BE THE SAME SIZE AND QTY AS THE VERTICAL REINFORCING (UN